

ABM ENGINE

PRINCIPLE 3 — COMPLETE RULES

DISTRIBUTED DECISION INTELLIGENCE ARCHITECTURE

All 104 Rules — 10 Deep Iterations — Full Explanations

Decimal Technologies | KSA Banking & Financial Institutions

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THE REWRITTEN PRINCIPLE

The ABM Engine is a distributed decision-making system. AI contributes memory, analytical scale, pattern recognition, consistency, and uncertainty reduction across the account universe. Humans contribute accountability, contextual judgment, political intelligence, relationship stewardship, ethical reasoning, and exception handling. The architecture does not seek to maximize automation. It seeks to maximize DECISION QUALITY under uncertainty while preserving trust, accountability, explainability, resilience, and human expertise.

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PRINCIPLE 3 — DISTRIBUTED DECISION INTELLIGENCE ARCHITECTURE

Original statement: 'AI assists humans, never replaces them.' After 10 iterations of 5-Whys analysis, this became: 'The system is a distributed decision-making architecture where AI provides scale, memory, consistency, and uncertainty reduction, while humans provide accountability, judgment, political intelligence, and relationship stewardship. The objective is not automation but superior decisions under uncertainty.'

INITIAL PASS: Human-AI Review Logic (4 Rules)

These 4 rules were discovered in the first 5-Whys pass across all principles.

NEW RULE [HUMAN-AUTO-001]

Three-phase human review reduction: Phase 1 (Days 0-90): 100% human review, no exceptions. Phase 2 (Days 91-180): Optional for P3/P4 + non-C-suite if AI first-pass QC rate >85% AND human override rate <10%. Phase 3 (Days 181+): Optional for P2 + Director-level if QC >92% AND override <5%. C-suite messages: ALWAYS human reviewed, permanently. WHY: C-suite errors are reputation-destroying. Lower-risk messages can be automated once trust is established through data.

NEW RULE [HUMAN-AUTO-002]

Escalation chain for message review: Hour 0-4: Primary reviewer (account_owner). Hour 4-8: Backup reviewer (designated buddy). Hour 8-24: Team lead / sales manager. Hour 24+: Message EXPIRES from review queue — must be regenerated with fresh context if a new send window opens. WHY: Context goes stale. A message drafted for a signal 48 hours ago may reference "this week's announcement" which is now last week's.

NEW RULE [HUMAN-AUTO-003]

Track edit distance on human edits: MINOR_TWEAK <15% changed, SIGNIFICANT_EDIT 15-50%, MAJOR_REWRITE >50%. If MAJOR_REWRITE rate exceeds 30% for a specific signal_type or persona_type, flag the AI prompts for that category as underperforming — trigger prompt optimization review. WHY: High rewrite rates mean the AI is failing for that category. Fix the prompts, not the individual messages.

NEW RULE [HUMAN-AUTO-004]

Humans can NEVER add factual claims that aren't in enrichment data without those claims being logged separately from AI-generated claims. The system is not responsible for validating human additions, but it must track them for audit. WHY: Mixing unverified human claims with AI claims contaminates the audit trail.

DEEP DRILL: Foundation & Context (5 Rules)

NEW RULE [AIHUMAN-FOUNDATION-001]

"The AI-Human Boundary Principle": AI operates in the domain of DATA (signals, scores, research, patterns, drafts). Humans operate in the domain of RELATIONSHIPS (trust, judgment, negotiation, reputation). The boundary between these domains shifts as AI proves reliability, but the RELATIONSHIP domain is permanently human. The system must be designed so AI never takes actions in the relationship domain without explicit human authorization. WHY: Enterprise banking sales in KSA are trust-based, relationship-based, and culturally nuanced. A single AI hallucination to a SEVP at SNB could permanently close the door. The cost of one bad message outweighs 100 automated good ones.

NEW RULE [AIHUMAN-FOUNDATION-002]

"The 90/10 Resource Model": AI handles approximately 90% of work volume (signal monitoring, scoring, enrichment, research, reasoning, drafting, QC, engagement tracking, analytics). Humans handle approximately 10% (message approval, relationship conversations, strategic overrides, handoff management, outcome classification). But the 10% human contribution carries 90% of the REPUTATIONAL RISK. Therefore: every human decision point must be presented with complete AI-prepared context so the human can make a high-quality judgment in minimum time. WHY: Without AI, a BD person spends 2-3 hours researching per account. With AI, they spend 5 minutes reviewing a strategically reasoned draft.

NEW RULE [AIHUMAN-CONTEXT-001]

"Human Decision Context Package" — Every human decision point must present complete context: (1) Message Approval: full reasoning chain, triggering signal with source, account summary, persona profile, chosen strategy with rationale, Golden Window status, conflict flags, ICS score, Effective Opportunity. (2) Sequence Strategy: contacts in order, channel reasoning, timing rationale, competitive landscape, relationship paths, committee coverage, blocker/veto flags. (3) Handoff Response: full engagement history, reply text with sentiment, meeting brief, talking points, likely objections, comparable wins. (4) Override Decision: system recommendation with rationale, what changes if overridden, historical override outcomes, risk assessment. (5) Outcome Classification: sequence timeline, engagement events, contact behavior, signal quality assessment, AI suggested classification. WHY: "Approve this? Yes/No" produces worse decisions than "Here's why we're sending this, here's the risk — approve?"

NEW RULE [AIHUMAN-DISAGREE-001]

"Disagreement Resolution Protocol": The human ALWAYS wins during MVP. But disagreement must be CAPTURED: (a) record AI output, human change, human stated reason (required, min 10 chars), (b) data feeds three mechanisms — prompt improvement (humans change ANGLE → persona prompts need updating), knowledge gap (humans add facts AI didn't know → enrichment incomplete), style mismatch (humans change tone/length → generation constraints need adjusting), (c) monthly report: "Top 10 reasons humans overrode AI" — single most valuable input for AI improvement, (d) CRITICAL: system must NEVER train AI to simply mimic human edits without understanding WHY. A human might edit for situation-specific reasons that shouldn't generalize.

NEW RULE [AIHUMAN-DISAGREE-002]

"AI Guardrails on Human Actions" — AI doesn't override humans but CAN warn: SOFT WARNINGS (overridable, logged): message contains unverifiable claims ("Source not found in enrichment"), edit introduces banned phrase or exceeds length, human force-promotes account with ICS <40. HARD BLOCKS (require escalation): message would send outside KSA business hours or during Ramadan/Eid, message targets DISQUALIFIED or EXHAUSTED contact, message to DORMANT account in cooling-off period. WHY: Humans make mistakes too — especially tired reviewers at 5pm Thursday.

DEEP DRILL: Explainability Layer (5 Rules)

NEW RULE [EXPLAIN-001]

"Reasoning Chain Transparency" — Every AI-generated message carries a "Reasoning Passport": SIGNAL: text, date, source, confidence. INTERPRETATION: what it means for buying intent (Agent 1). ACCOUNT CONTEXT: organization summary (Agent 2). PERSONA ANALYSIS: role motivations, decision criteria (Agent 3). PAIN INFERENCE: likely operational pains with reasoning (Agent 4). STRATEGY: chosen angle, value proposition, CTA rationale (Agent 5). QC RESULT: all checks passed, confidence score (Agent 7). Stored permanently alongside message. Visible to reviewer. Part of handoff package. WHY: If you can't explain why you sent a message, you shouldn't send it.

NEW RULE [EXPLAIN-002]

"Score Decomposition" — Every score breakable into contributing factors: Dynamic Score 82 = Signal Strength 75 (weight 35%, contribution 26.25) + Regulatory Pressure 90 (30%, 27) + Reachability 70 (20%, 14) + Warmth 60 (15%, 9) + intent multiplier + criticality bonus. ICS decomposed similarly. Effective Opportunity shows each multiplier. Every number has traceable source. No black boxes. WHY: When human asks "why is this account #3 not #1?" system shows exactly which dimensions differ.

NEW RULE [EXPLAIN-003]

"Negative Decision Logging" — Every time system decides NOT to act, it logs: "Account X not promoted to HOT because ICS = 35 (below 40 per GOV-001)." "Message for Contact Y blocked: OUTREACH_EXHAUSTED (per FAIL-FATIGUE-002)." "Outreach deferred: Golden Window expired, needs regeneration (per TIMING-WIN-002)." These logs are queryable. Owner asks "Why hasn't anything happened with Bank X?" and gets a specific answer. WHY: Silent inaction is the hardest thing to debug. Negative decisions must be as visible as positive ones.

NEW RULE [EXPLAIN-004]

"Human Decision Audit Trail" — Every human action logged with: WHO (which human), WHAT (approved/edited/rejected/override/escalated), WHEN (timestamp), WHY (stated reason — required for edits/rejections/overrides), AGAINST (what AI recommended if different), OUTCOME (what happened as result). Creates complete audit trail where BOTH AI reasoning and human judgment are visible. WHY: If a message causes a problem, the trail shows: AI recommended X, human changed to Y, for reason Z. Accountability is clear.

NEW RULE [EXPLAIN-005]

"Accountability Framework" — 4 layers: Layer 1 — AI QC Agent: accountable for catching hallucinations, tone violations, factual errors. Layer 2 — Human Reviewer: accountable for strategic judgment (right angle? right timing? right tone for relationship?). Layer 3 — Account Owner: accountable for relationship outcomes. Layer 4 — System Designer (BD leadership): accountable for rules, thresholds, workflows, guardrails. Post-mortem identifies WHICH layer failed. No finger-pointing between AI and humans. WHY: Clear accountability prevents blame-shifting and drives targeted fixes.

DEEP DRILL: AI Behavioral Boundaries (3 Rules)

NEW RULE [AIBOUNDARY-001]

"AI Permission Matrix" — Exhaustive CAN/CANNOT list: CAN: gather public info, calculate scores, generate drafts, tailor messaging, recommend strategy, classify signals/sentiment/outcomes, analyze performance, validate outputs, propose send times, prepare CRM data. CANNOT: send messages without approval (MVP), change scoring formulas/weights/thresholds, reclassify human-verified items, override its own QC rejections, override business hours/holidays, create/modify/delete CRM deals, access private systems, execute strategy changes without confirmation. DEFAULT for unlisted actions: AI CANNOT. WHY: Without explicit boundaries, AI scope creeps.

NEW RULE [AIBOUNDARY-002]

"Approved Claims Library" — AI can ONLY make claims about Decimal from a pre-approved library: verified client references (REFERENCEABLE only), specific product capabilities (reviewed by product team), performance metrics (verified by data), compliance certifications (verified by legal), integration capabilities (verified by engineering). Claims not in library = "CLAIM_NOT_VERIFIED" flag + placeholder. AI must NEVER: guarantee outcomes ("WILL increase lending 40%"), make comparative claims ("better than Temenos"), claim non-existent capabilities, reference NDA-protected info. WHY: One false capability claim to a bank CTO creates compliance and trust crisis.

NEW RULE [AIBOUNDARY-003]

"Cultural Sensitivity Rules" — KSA-specific norms hardcoded: FORMALITY: formal titles always in initial outreach, never initiate informality. RELIGION: never reference practices/beliefs unless contextually appropriate (Ramadan greetings OK during Ramadan). GENDER: no assumptions about capabilities based on gender, same professional tone regardless. HIERARCHY: never suggest lower-ranking should "convince" superiors — frame as "sharing information." NATIONAL PRIDE: frame Decimal as supporting Vision 2030, never as "teaching Saudi banks." COMPETITOR RESPECT: respectful tone even when positioning against competitors. WHY: Cultural missteps in KSA banking are relationship-ending.

DEEP DRILL: Failure Modes 1-6 (11 Rules)

FM1 — Rubber-Stamping (2 Rules)

NEW RULE [FAIL-RUBBER-001]

"Anti-Rubber-Stamping Mechanisms": (a) Minimum review time tracking: <30 seconds for email or <15 for LinkedIn = "SPEED_REVIEW" flag (meaningful review takes 60-90 seconds). (b) Spot-check canaries: weekly, system inserts deliberately flawed message (wrong company name, non-existent signal). Approved canary = "CANARY_MISSED" alert. Flaw caught before sending. Reviewer notified. (c) Review Quality Score per reviewer: canary catches, edit rate, time per review, agreement with post-hoc audits. (d) Rotation: reviewers don't review same account repeatedly. Monthly rotation. WHY: After 50 good messages, reviewer assumes #51 is fine. Canaries test vigilance.

NEW RULE [FAIL-RUBBER-002]

"Review Quality Culture": Canary system framed as QUALITY TOOL not surveillance: (a) all reviewers told upfront during training, (b) results private, not shared with team, (c) missed canary = coaching not discipline, (d) Review Quality Score for training needs, not performance evaluation, (e) canary frequency decreases as reviewer quality improves. WHY: Surveillance-framed systems create resentment and gaming. Quality-framed systems create improvement.

FM2 — AI Hallucination (2 Rules)

NEW RULE [FAIL-HALLUC-001]

"Three-Layer Hallucination Prevention": Layer 1 — Generation Constraint: Agent 6 explicitly instructed to use ONLY facts from enrichment, signals, and Approved Claims Library. Unverifiable = "CLAIM_UNVERIFIED" placeholder. Layer 2 — Automated Fact-Check: before Agent 7, cross-reference every entity name, number, date, claim against databases. Unmatched claims flagged with specific text highlighted. Layer 3 — Human Review: flagged claims shown in RED. Unflagged in normal text. Visual hierarchy focuses human attention on highest-risk parts. WHY: Triple-layer means hallucination must fool generator, fact-checker, AND human.

NEW RULE [FAIL-HALLUC-002]

"Hallucination Incident Response": If hallucinated claim IS sent: Step 1: account owner notified within 1 hour. Step 2: assess — did prospect notice? Step 3: if noticed, owner contacts personally, acknowledges error honestly, provides correct info. Step 4: incident logged as "HALLUCINATION_BREACH" with which layer(s) failed + root cause. Step 5: add hallucination pattern to fact-checker negative rules. WHY: Honest correction builds more trust than silence.

FM3 — AI Staleness (2 Rules)

NEW RULE [FAIL-STALE-001]

"AI Currency Maintenance Schedule": Approved Claims Library: monthly (triggered by product launch, client win, certification, pricing change). Persona Prompts: quarterly (triggered by persona mismatch feedback). Strategy Prompts: monthly (triggered by market/competitive changes). Cultural Rules: quarterly (triggered by policy changes, lessons learned). Competitive Intelligence: continuously. Signal Classification: monthly (triggered by false positive thresholds). QC Criteria: quarterly (triggered by systematic failures). All version-controlled. Rollback within 1 hour if degradation. WHY: AI that doesn't evolve becomes irrelevant as markets change.

NEW RULE [FAIL-STALE-002]

"AI Performance Degradation Detection": Track monthly — thresholds for "STALE_AI" trigger: QC approval rate drops 10+ points from 90-day average, human override rate increases 10+ points, reply rate declines 30%+ (controlling for seasonality), average edit distance increasing, MAJOR_REWRITE rate >20% for any category. STALE_AI triggered: emergency prompt review, enrichment audit, competitive check, claims library verification. Fix within 7 business days. WHY: Gradual degradation is invisible reviewing individual messages. Systematic measurement catches decline.

FM4 — Human Over-Reliance (1 Rule)

NEW RULE [FAIL-OVERRELY-001]

"Human Independent Judgment Preservation": (a) Monthly "Red Team": owners independently rank top 5 accounts WITHOUT seeing system ranking first, then compare. Divergence is GOOD. Perfect alignment after 6+ months = possible over-reliance. (b) AI Confidence Display: always show confidence level with recommendations. Never present as certainties. (c) Override encouragement: if owner NEVER overrides in 90 days, manager conversation: genuinely agreeing or defaulting? (d) Market Sensing Reports: monthly BD team report based on THEIR conversations, independent of AI analysis. Compare "what market tells us" vs "what AI tells us." WHY: An AI-dependent team cannot

function during an outage.

FM5 — Wrong Metric Optimization (2 Rules)

NEW RULE [FAIL-WRONGMETRIC-001]

"Metric Hierarchy": NORTH STAR: pipeline revenue from ABM-sourced opportunities. PRIMARY: meetings booked, positive reply rate. SECONDARY: total reply rate, signal-to-meeting conversion. OPERATIONAL: open rate, click rate, LinkedIn accept rate, QC pass rate. AI must NEVER optimize operational at expense of primary. A change that increases open rates but decreases positive reply rates is BAD. Feedback loops ONLY recalibrate based on primary/north star. Operational metrics for diagnostics only. WHY: Optimizing vanity metrics produces impressive dashboards and empty pipeline.

NEW RULE [FAIL-WRONGMETRIC-002]

"Metric Manipulation Detection": Divergence between operational and primary = warning: open rate up but reply flat = misleading subjects. QC pass up but override also up = QC weakened. LinkedIn accept up but DM reply flat = connection good, follow-up bad. Monthly check required. WHY: Metrics that improve independently of outcomes indicate gaming or miscalibration.

FM6 — AI Memory Leakage (2 Rules)

NEW RULE [FAIL-LEAK-001]

"Account Isolation Principle": Each reasoning chain receives ONLY that account's data: enrichment, signals, contact personas, plus universal Approved Claims Library and general strategy guidelines. Must NOT receive other accounts' data, engagement outcomes, or response patterns. WHY: What works at SNB may fail catastrophically at Al Rajhi. Each bank has its own culture, priorities, and politics.

NEW RULE [FAIL-LEAK-002]

"Cross-Account Learning Rules": Patterns CAN be learned across accounts if: (a) derived from 5+ accounts, (b) about PERSONA TYPES not specific people, (c) about MESSAGE CHARACTERISTICS not specific content, (d) applied as TENDENCY not hard rule. ALLOWED: "CTOs tend to prefer messages under 100 words." NOT ALLOWED: "CTO at Bank A liked compliance pitch — use for Bank B." WHY: Aggregate behavioral patterns are useful. Specific account strategies are not transferable.

DEEP DRILL: Trust Progression Model (1 Rule)

NEW RULE [TRUST-001]

"Trust Progression Framework" — 6 levels: ZERO TRUST (Day 0): AI drafts only, cannot queue. 100% review. Advance after 30 days no incidents. INITIAL TRUST (Day 30+): AI queues for review. Auto-enrichment, auto-scoring. QC >60% to advance. DEVELOPING TRUST (Day 60+): QC >80%. Review SLA can extend to 8h for P3/P4. ESTABLISHED TRUST (Day 90+): QC >85%, override <10%. P3/P4 non-C-suite can send after AI QC only. P1/P2/C-suite: 100% human. MATURE TRUST (Day 180+): QC >92%, override <5%. P2 Director optional review. C-suite ALWAYS human. FULL PARTNERSHIP (Day 360+): AI handles routine. Humans focus on relationships/strategy. C-suite/P1 always human. Quarterly full audit. CRITICAL: Trust can REGRESS. Single hallucination = drop one level. Reputation damage = drop to INITIAL. WHY: Trust earned incrementally, lost instantly — mirrors human trust in enterprise relationships.

ITERATION 1 — AI LEARNING & EVOLUTION ARCHITECTURE

How does the AI actually get smarter without getting dumber? Learning in a high-stakes enterprise system is itself dangerous — learning the WRONG lessons can be worse than not learning at all.

NEW RULE [AILEARN-001]

"Minimum Sample Size for Learning" — No parameter changed with fewer than: Scoring weights: 10 meetings across 3+ signal types. Persona prompts: 5 completed sequences to that persona type. Signal prioritization: 15 signals of that type with tracked outcomes. Timing rules: 20 messages in proposed window. QC thresholds: 30 messages through pipeline. Sequence structure: 10 completed sequences. Below minimum = logged as OBSERVATION, not acted upon. Observations accumulate until threshold met. WHY: Overfitting to 3 data points is worse than not learning. "CTO at Bank X liked compliance" is anecdote. "7 of 10 CTOs at Islamic banks responded to compliance" is pattern.

NEW RULE [AILEARN-002]

"Learning-to-Change Pipeline" — 7 stages: Stage 1 OBSERVATION: AI detects pattern. Stage 2 HYPOTHESIS: formulated as testable statement. Stage 3 VALIDATION: check sample size, confounding variables, survivorship bias. Stage 4 PROPOSAL: specific parameter change + expected impact + rollback criteria. Stage 5 HUMAN APPROVAL: BD leadership reviews. No change without sign-off. Stage 6 A/B DEPLOYMENT: 30% gets new approach, 70% existing. Compare 2+ weeks. Stage 7 FULL DEPLOYMENT or ROLLBACK: improvement → deploy 100%. Neutral/negative → rollback. WHY: Prevents reckless changes (too fast, too little data) and stagnation (no process to change).

NEW RULE [AILEARN-003]

"Prompt Version Control": Every prompt carries: version_id (semantic major.minor.patch), effective_date, change_description, change_rationale (linked to AILEARN-002 proposal), performance_baseline. Every AI_GENERATION record stores prompt_version_id. Enables: "In March, v2.3.1 produced 4.2% reply rate. April v2.4.0 produced 5.8%." Or: "v3.0.0 doubled hallucination rate, rolled back to v2.4.0 in 3 days." No prompt change without version increment. Direct database editing without versioning = SYSTEM VIOLATION. WHY: Without version control, you can't correlate AI quality changes with the prompt changes that caused them.

NEW RULE [AILEARN-004]

"Chain Coherence Testing": When ANY agent prompt modified, full 7-agent chain regression-tested: run against 10 historical accounts with known outcomes. Compare to original output, actual sent message, and human-edited version. Pass: equal or better on 7+ of 10 (human-judged, not automated). Fail: worse on 4+ → reject change, investigate downstream effects. WHY: The chain is a system, not independent components. Testing in isolation misses interaction effects.

NEW RULE [AILEARN-005]

"AI Capability Gap Registry": Maintained backlog of things AI cannot do well but should: Arabic message generation (HIGH priority, 6mo target), competitive positioning when competitor unknown (MEDIUM, 9mo), optimal send time prediction (LOW, 18mo), Arabic news signal detection (HIGH, 6mo), organizational politics from public data (LOW, likely never fully automatable). Reviewed quarterly. Top gaps get engineering resources. WHY: Knowing what you CAN'T do is as important as improving what you CAN do.

NEW RULE [AIMODEL-001]

"Model Change Protocol": Step 1 EVALUATION: new model against 10 historical test cases. Step 2 SHADOW MODE: 2 weeks parallel with existing model (only existing used for outreach, new logged for comparison). Step 3 COMPARATIVE ANALYSIS: QC rates, output length, hallucination flags, blind human side-by-side. Step 4 CUTOVER: equal or better → gradual 30%→60%→100% over 2 weeks. Worse on any critical dimension → reject. Step 5 MONITORING: 30-day enhanced monitoring. WHY: "AI got worse" is extremely hard to diagnose after the fact. Shadow mode prevents production quality drops.

NEW RULE [AIMODEL-002]

"Model Fallback Architecture": Always TWO operational models: PRIMARY (main) and FALLBACK (tested, proven, ready to deploy). Fallback activation: automatic for availability (primary down 30+ min), manual for quality (QC rejection spikes above 40%, latency >30s consistently — requires human confirmation of actual degradation). WHY: Single-model dependency is a catastrophic risk. Outage at the model provider = complete system stoppage without fallback.

NEW RULE [AIMODEL-003]

"Model Cost Management": Track per-operation costs: signal classification, account research, full 7-agent chain, QC validation. Monthly budget ceiling defined by leadership. Costs at 80% of ceiling → alert. Exceed ceiling → essential-only (P1/P2 + HOT accounts; P3/P4 + WARM deferred). Optimization: cheaper model for low-stakes ops, cache enrichment intelligence, batch P3/P4 classifications. WHY: AI costs scale with volume. Surge month without tracking can blow budget.

NEW RULE [AIQUALITY-001]

"Quality Degradation Early Warning" — 6 indicators tracked monthly: (1) Personalization depth: avg account-specific sentences/message, threshold drops below 2.0 from ~3.5 baseline. (2) Reasoning chain length: avg word count of agent outputs, threshold 30%+ below baseline. (3) Vocabulary diversity: unique/total words per week, threshold below 0.4 from ~0.55. (4) Signal reference specificity: % referencing specific signal, threshold below 70% from ~90%. (5) CTA variation: unique CTAs per month, threshold below 4 from ~8. (6) Strategy diversity: unique angles per month, threshold below 3. Any threshold crossed → "QUALITY_DRIFT" alert. Fix investigated within 7 days. WHY: Quality degradation is invisible reviewing individual messages. Only systematic measurement catches gradual decline.

NEW RULE [AIQUALITY-002]

"Gold Standard Library": 20 exemplary messages that were sent, received positive replies, led to meetings. Monthly: generate 5 new messages for SAME accounts/contacts using current AI. Human evaluator scores: BETTER/EQUAL/WORSE vs gold standard on personalization, strategic soundness, writing quality. >40% rated WORSE → quality degraded, trigger full AI review. Library updated quarterly: add new excellent messages, retire outdated ones. WHY: Benchmarking against your own best work prevents drift that no metric catches.

ITERATION 2 — HUMAN CAPABILITY & KNOWLEDGE LAYER

What happens to the HUMANS over time? The AI reshapes how they work, think, and develop skills. If humans atrophy, the entire system becomes fragile.

NEW RULE [HUMANSKILL-001]

"Manual Capability Preservation": Every system operator must demonstrate WITHOUT AI: (a) research account from public sources, write 1-page intelligence brief, (b) write personalized outreach email from scratch, (c) manually calculate account score, (d) prioritize 10 accounts for outreach, (e) conduct meeting preparation without AI brief. Skills assessment: every 6 months, real accounts. WHY: AI is a capability AMPLIFIER. If humans can't function independently, the amplifier produces zero when it breaks.

NEW RULE [HUMANSKILL-002]

"Manual-First Onboarding": Week 1: operate manually — research without AI summaries, write without AI drafts, score by hand. Week 2: shadow AI — see AI outputs, compare to manual work, understand strengths/weaknesses. Week 3: operate WITH AI — review, approve, manage accounts. Week 4: certified reviewer. Skip weeks 1-2 for experienced enterprise sales hires, but certification still required. WHY: Must understand what AI does before judging whether it does it well.

NEW RULE [HUMANSKILL-003]

"Knowledge Externalization": Critical intelligence must NOT exist only in AI system: (a) account summaries exportable to PDF/DOCX, (b) contact data synced to CRM (HubSpot is backup), (c) signal history exportable to CSV/JSON, (d) scoring history exportable, (e) prompt library in version-controlled docs, (f) Approved Claims Library as standalone document. Monthly backup verification: can we reconstruct top 10 accounts from external backups alone? If NO → externalization insufficient. WHY: AI systems aren't permanent. Organizational knowledge must survive any single system failure.

NEW RULE [HUMANFATIGUE-001]

"Decision Load Balancing": (a) Max 10 sequential decisions without break — system prompts 5-minute break reminder. (b) Highest-stakes decisions in morning (9-11am): P1, C-suite, handoffs. P3/P4 in afternoon. (c) Complexity variation: don't present 10 conflicted accounts in a row. (d) Queue visibility: reviewer always sees queue depth. Surge = load balancing flag. WHY: Decision quality declines measurably throughout the day. System design should account for human biology.

NEW RULE [HUMANFATIGUE-002]

"Decision Quality Correlation": Track: time of day vs approval rate (98% after 3pm but 82% before noon → afternoon rubber-stamping), queue depth vs review time (drops when deep → rushing), day of week vs override rate (Friday quality drop → end-of-week fatigue). Used to ADJUST SYSTEM not penalize humans: shift P1/P2 to mornings, add afternoon reviewer, reduce Friday load. WHY: Optimizing the human is as important as optimizing the AI.

NEW RULE [AITRANSAPRENCY-001]

"AI Literacy Program": Level 1 (ALL users): what AI does and doesn't do. Level 2 (Reviewers + Owners): how 7-agent chain works, how to read Reasoning Passport, how to identify AI errors, common failure patterns. Level 3 (BD Leadership): scoring formula, three-score model, feedback loops, analytics interpretation, AI change approval. Assessment required before system operation. Annual refresher. WHY: Users who don't understand the AI can't evaluate its output.

NEW RULE [AITRANSAPRENCY-002]

"AI Limitation Visibility": System proactively surfaces limitations where decisions are made: (a) score with missing data: "Based on incomplete data. Digital maturity: unknown." (b) message with low-confidence steps: "Pain inference confidence: LOW. Consider accuracy." (c) recommendation with close call: "Recommended: CTO first (65%). Alternative: CDO (55%). Close call — your judgment should decide." WHY: Presenting output without uncertainty markers encourages blind trust.

NEW RULE [HUMANMEMORY-001]

"Account Strategy Notes": Every owner maintains written notes per HOT account, updated monthly: current strategic assessment (own words, not AI), learnings from direct conversations, relationship dynamics observed, decision rationale for overrides/delays, prediction of what happens next and why. Stored alongside AI intelligence. Part of handoff package. Reviewed in quarterly strategy sessions. WHY: When owner leaves, human intelligence about politics and dynamics leaves too — unless written down.

NEW RULE [HUMANMEMORY-002]

"Post-Meeting Intelligence Capture": Within 24 hours of every meeting/significant conversation, owner logs: topics discussed, prospect's stated priorities and concerns, political dynamics observed, next steps agreed, owner's deal assessment, budget signals, competitive signals. Feeds into: enrichment updates, consensus scoring, risk assessment, AI context for future messages. WHY: A 30-minute meeting generates more intelligence than 6 months of signal monitoring.

ITERATION 3 — ADVERSARIAL, CATASTROPHIC & ETHICAL LAYER

What happens when things go catastrophically wrong? Not 'slightly generic message' but 'reputation crisis,' 'system being gamed,' or 'ethical dilemma.'

NEW RULE [ADVERSARIAL-001]

"Input Manipulation Detection" — 5 attack vectors: (1) Fake signal injection: fake news about bank triggers outreach → mitigated by Source Reliability + Observability Score + two-stage classification. (2) Engagement inflation: competitor/troll opens all emails → detect bot patterns (every email opened within 30s), flag SUSPICIOUS_ENGAGEMENT, don't count for scoring. (3) LinkedIn honeypot: fake profile mimicking executive → cross-reference with Apollo/Clearbit, UNVERIFIED_IDENTITY blocks outreach until verified. (4) Competitor intelligence: competitor responds to learn our pitch → never reveal pricing/strategy in automated messages, only through human conversations. (5) System probing: testing how system responds → rate-limit signal ingestion per source, flag unusual patterns. WHY: Any system trusting external data can be manipulated through that data.

NEW RULE [ADVERSARIAL-002]

"Internal Gaming Detection": (a) Phantom meetings: owner marks low-quality as "booked" → cross-reference CRM calendar + follow-up activity. (b) Speed-approval: reviewer approves without reading → FAIL-RUBBER-001 mechanisms. (c) False manual signals: BD enters fake signals → random 20% monthly audit, signals never leading to activity are suspicious. (d) Suppressive overrides: owner suppresses accounts → override frequency tracking, persistent overrides without reason = review. Response: findings for PROCESS IMPROVEMENT not employee punishment. If incentives encourage gaming, fix incentives. WHY: Internal gaming produces beautiful dashboards and zero pipeline.

NEW RULE [CATASTROPHE-001]

"Catastrophic Failure Classification": SEV-1 MARKET-THREATENING: grossly offensive message to senior executive, data breach exposing contacts, 500 messages sent due to bug. Response: KILL SWITCH within 5 min, CEO notified, external communication within 24h, post-mortem within 72h. SEV-2 ACCOUNT-THREATENING: hallucination to C-suite despite QC+human, same person contacted from 3 sequences, confidential info leaked cross-account. Response: pause affected account, owner contacts within 4h, honest acknowledgment, post-mortem within 1 week. SEV-3 OPERATIONAL: AI garbage for 2 hours unnoticed, scoring miscalculation for batch, enrichment returns wrong data. Response: scope impact within 2h, fix within 24h, audit failure-window outputs. SEV-4 MINOR: single typo, wrong formatting, one account scored incorrectly with no outreach. Response: fix and document.

NEW RULE [CATASTROPHE-002]

"Kill Switch": One-click emergency stop available to BD leadership + system admin: (a) halts ALL outreach (email + LinkedIn), (b) pauses all pending messages in queue, (c) stops all AI generation, (d) continues signal ingestion and scoring (intelligence doesn't stop), (e) alerts all operators: "SYSTEM HALTED by [who] at [time]. Reason: [required]." Activation requires reason. Deactivation requires BD leadership approval + root cause resolved. WHY: Every minute of continued operation during catastrophe makes it worse.

NEW RULE [CATASTROPHE-003]

"Post-Mortem Framework" — Every SEV-1/SEV-2: WHAT happened (exact events + timestamps). WHO affected (accounts, contacts, messages). WHY it happened (root cause — which layer failed). WHY NOT CAUGHT (which safety layers should have prevented and didn't). HOW TO PREVENT (specific changes, assigned owner, deadline). BUSINESS IMPACT (relationship damage, recovery possibility). Completed within 7 days. Findings shared with team. Changes within 30 days. Repeat items from previous post-mortems = escalation. WHY: Structured learning from failures prevents recurrence.

NEW RULE [ETHICS-001]

"Ethical Absolutes — What System Must Never Do": (a) NEVER exploit personal vulnerability — mark contact "PERSONAL_SENSITIVITY" if intelligence reveals personal difficulty. (b) NEVER misrepresent identity — messages sent on behalf of named Decima employees, no fake personas. (c) NEVER manufacture urgency — reference REAL urgency only, no "offer expires Friday" when it doesn't. (d) NEVER disparage competitors by name in automated outreach. (e) NEVER use engagement data punitively — don't escalate to boss or cc colleagues. (f) NEVER ignore opt-out — ABSOLUTE and PERMANENT unless explicitly reversed by contact. WHY: The system manipulates human attention. That power requires ethical constraints.

NEW RULE [ETHICS-002]

"Data Ethics": (a) Proportional collection: only data serving defined business purpose. (b) Data minimization in prompts: agents receive only needed data. (c) Retention limits: ARCHIVE accounts 12+ months → purge/anonymize personal contact details. Organizational intelligence retained. (d) Transparency on request: if contact asks "how did you get my info?" answer honestly. (e) PDPL compliance: quarterly review against Saudi data protection requirements. WHY: Legal compliance and moral responsibility for personal data handling.

NEW RULE [ETHICS-003]

"Personalization vs Manipulation Line": ACCEPTABLE: "Given your focus on digital lending growth..." (real business priority). "Banks that modernized saw 40% improvement..." (factual, relevant). UNACCEPTABLE: "Your competitors are already ahead" (manufactured fear, may not be true). "You can't afford to wait" (pressure not grounded in deadline). "As a CTO who cares about their legacy..." (identity manipulation). "Other executives at your bank are interested" (internal social proof, may be false and politically dangerous). THE TEST: "Would recipient feel RESPECTED seeing the AI reasoning?" If they'd feel analyzed/profiled/manipulated → crosses the line. Agent 4 produces BUSINESS pains, never insecurities. Agent 5 selects VALUE angles, never fear. Agent 6 writes with RESPECT, never urgency manufacturing.

NEW RULE [GOVERNANCE-001]

"System Governance Structure": Three layers: OPERATIONAL (Daily): owners + reviewers — message quality, engagement, day-to-day. Report to BD leadership. STRATEGIC (Monthly): BD leadership — portfolio performance, AI quality, learning pipeline, competitive landscape, capacity. Decides AI changes, account universe, resources. ETHICAL (Quarterly): BD leadership + legal + external advisor — system behavior vs ethical boundaries, PDPL, data practices, AI transparency, incidents. WHY: Different oversight frequencies for different risk horizons.

NEW RULE [GOVERNANCE-002]

"Quarterly System Audit": 6 dimensions: (a) AI quality: QC rates, hallucination incidents, override patterns. (b) Outreach ethics: complaints, opt-out compliance, data handling. (c) Human operations: reviewer quality, fatigue indicators, skill preservation. (d) System integrity: kill switch TESTED (yes, actually test quarterly), backup restoration verified, data externalization current. (e) Financial: AI costs vs budget, cost per opportunity, ROI. (f) Competitive: market position changes, competitor movements, reference pipeline. Results documented. Action items tracked. Repeat items = escalation. WHY: Comprehensive quarterly health check prevents gradual system degradation.

NEW RULE [GOVERNANCE-003]

"Bias Monitoring": Quarterly analysis: (a) Gender: response rates by contact gender significantly different? Messaging inadvertently gendered? (b) Segment: systematically favoring certain bank types? Islamic banks getting less attention despite similar fit? (c) Seniority: appropriately distributing outreach or always defaulting to C-suite? (d) Recency: overly weighting recent signals while ignoring established accounts? Statistically significant bias detected → investigate root cause (weights? prompts? enrichment?) and correct. WHY: Undetected bias in an automated system scales bias to every outreach the system sends.

NEW RULE [GOVERNANCE-004]

"Cross-Principle Consistency Check": Quarterly, verify that rules from Principle 3 don't conflict with Principles 1, 2, 4, or 5. As the rule set grows, contradiction risk increases. Example: Principle 1 says "respond immediately to P1 signals" but Principle 3 says "every message needs 4-hour human review" — what happens when P1 arrives at 10pm? Resolution: processing is immediate (P1), human review happens next business morning (P3), message is in Ready-to-Fire queue by Sunday 9am. Both principles satisfied through queue architecture. Document all such tensions and their resolutions. WHY: Principles that contradict each other in practice produce unpredictable system behavior.

ITERATION 4 — TRUST ECONOMICS LAYER

Trust is not binary. Trust is an economic asset. It accumulates through deposits (successful outcomes) and depletes through withdrawals (failures). The withdrawal rate is always faster than the deposit rate. Trust is not error rate — it is consequence-weighted reliability.

NEW RULE [AITRUST-001]

"Trust Capital Ledger" — Each AI subsystem maintains Trust Capital (0-100, starting at 50): Signal Classification: +1 per correct, -5 per false positive triggering wasted outreach, -15 per false positive reaching prospect. Scoring: +2 per correct tier prediction, -3 per significant misrank, -10 per missed deal. Message Generation: +1 per no-edit approval, +3 per positive reply, -2 per MAJOR_REWRITE, -10 per hallucination passing QC, -25 per hallucination reaching prospect. QC: +1 per correct catch, +5 per caught hallucination, -5 per hallucination passing QC but caught by human, -20 per reaching prospect. Enrichment: +1 per verified-accurate data, -3 per wrong contact, -8 per wrong company data. Persona/Pain: +2 per confirmed-relevant pain, -3 per "that's not our priority" response. Thresholds: 70-100 TRUSTED (standard oversight). 40-69 CAUTIOUS (increased review). <40 DISTRUSTED (double review, root cause investigation). Asymmetric: builds slowly, drops fast. WHY: Trust built in drops, lost in buckets.

NEW RULE [AITRUST-001a]

"Trust Capital Gates Progression": To advance to ESTABLISHED TRUST: ALL subsystems ≥ 60 . To MATURE: ALL ≥ 70 . To FULL PARTNERSHIP: ALL ≥ 80 . ANY subsystem drops below 40 → system-wide trust regresses one level, cannot advance until recovery above 50. WHY: Excellent message generation with terrible signal classification = sending good messages to wrong accounts. Trust must be holistic.

NEW RULE [AITRUST-002]

"Override Severity Classification": COSMETIC (formatting, punctuation): 0 trust impact, 0 learning value. STYLISTIC (tone, formality): 0 trust impact, track as style preference. TACTICAL (swap case study, change CTA): -1 message generation trust, moderate learning value. STRATEGIC (change outreach angle, pain inference wrong): -3 strategy trust, -2 pain inference trust, HIGH learning value. SAFETY (block hallucination, prevent policy violation): -10 relevant subsystem, -5 QC if should have caught, CRITICAL learning value. Healthy monthly distribution: 60%+ cosmetic/stylistic, 25% tactical, 10% strategic, <5% safety. Safety >5% → AI quality system failing. Strategic >25% → strategic reasoning needs recalibration.

NEW RULE [AITRUST-003]

"Recommendation Verifiability Score": FULLY VERIFIABLE (90-100): all claims link to checkable sources. MOSTLY (60-89): most verifiable, some inferences. PARTIALLY (30-59): significant inferences, limited sources. LOW (<30): mostly inference-based. Displayed alongside messages. Low verifiability = reviewer applies MORE independent judgment. High = can rely on AI with spot-checking. WHY: Telling a reviewer "this is well-researched" without showing HOW well-researched asks for blind trust.

NEW RULE [AITRUST-004]

"Per-Reviewer Trust Calibration": Track per reviewer: total reviews, override rate by type, canary catch rate, average review time, correlation between approvals and outcomes. High-experience reviewer (200+, good canary, overrides improve outcomes) → expanded autonomy: larger batches, extended SLAs for low-risk. Low-experience/declining → tighter controls: smaller batches, shorter SLAs, spot-checks. WHY: Reward demonstrated judgment with efficiency. Support struggling reviewers.

NEW RULE [AITRUST-005]

"Failure Predictability Classification": EXPECTED FAILURE (system failed in known limitation scenario): x1 trust impact — confirms we understand limits. UNEXPECTED FAILURE (system failed where it should have worked): x5 trust impact — our reliability model is wrong. BLACK SWAN (entirely novel failure mode): x3 trust impact — serious but recoverable, expands understanding, triggers new prevention rule. WHY: A system that fails predictably is TRUSTWORTHY. A system that fails unpredictably is DANGEROUS.

ITERATION 5 — DECISION SCIENCE LAYER

How does a human actually decide? Most systems never ask. The same message can be approved by one reviewer and rejected by another. Understanding WHY creates the deepest learning.

NEW RULE [HUMANDEC-001]

"Decision Rationale Taxonomy" — Every significant decision (edit, reject, override) tagged with category: TIMING_CONCERN, RELATIONSHIP_CONCERN, SIGNAL_QUALITY_CONCERN, STRATEGIC_CONCERN, TONE_CONCERN, COMPLIANCE_CONCERN, FACTUAL_CONCERN, COMPETITIVE_CONCERN, POLITICAL_CONCERN, INTUITION. The INTUITION category is most valuable — expert pattern recognition below conscious awareness. Every INTUITION override followed up within 30 days: was instinct correct? If so, what was the underlying reason? This discovers NEW rules. Monthly analysis: dominant categories reveal systemic gaps.

NEW RULE [HUMANDEC-002]

"Decision Pattern Library" — After 6+ months with rationale capture, extract patterns: "WHEN [conditions], EXPERIENCED REVIEWERS tend to [decision], BECAUSE [reasoning]." Examples: "WHEN new CTO + hiring + Q1 → ACCELERATE, because fresh budget + receptive leader." "WHEN active M&A; → DELAY, because decisions freeze during integration." "WHEN connector confirms but no public signal → PRIORITIZE over public-signal-only accounts, because dark intent with validation converts better." Become: training material, AI strategy suggestions, institutional knowledge. NEVER automate — advisory only, human judgment applies. WHY: Institutionalizes expert judgment that otherwise leaves when experts leave.

NEW RULE [HUMANDEC-003]

"Intra-Reviewer Consistency": Monthly, present reviewer with PREVIOUSLY REVIEWED message (different presentation so unrecognized). Compare decisions. >85% consistent = stable reliable judgment. 70-84% = some drift, investigate categories. <70% = significant drift — unclear criteria, fatigue, or ambiguous review standards. Remediation: clarify criteria with examples, reduce load, pair with high-consistency reviewer for calibration. WHY: Same reviewer making different decisions on same message on different days adds noise not quality.

NEW RULE [HUMANDEC-004]

"Confidence-Outcome Tracking": For strategic decisions, capture reviewer confidence: HIGH/MODERATE/LOW/UNCERTAIN. Track outcomes: HIGH+Correct = well-calibrated (reward autonomy). HIGH+Wrong = overconfidence (MOST DANGEROUS — investigate false confidence source, share as learning). LOW+Correct = underconfidence (identify hesitation cause — data gaps or personality?). LOW+Wrong = appropriate uncertainty (system should provide more context). Monthly calibration analysis per reviewer. WHY: Confident and wrong is more dangerous than uncertain and right.

NEW RULE [HUMANDEC-005]

"Review Value Assessment": Post-hoc evaluation of every review: VALUE_ADDED (human changed output AND outcome improved), NO_VALUE (approved without changes, outcome same as AI default), NEGATIVE_VALUE (human changed AND outcome worse — rare but track), BLOCKED_HARM (human prevented damage — HIGHEST VALUE). Quarterly: what % add value? <20% for a message category → candidate for reduced review per trust progression. WHY: Review that doesn't change outcomes is cost without benefit. System earns right to reduced review by demonstrating human intervention rarely matters in specific categories.

ITERATION 6 — META-GOVERNANCE LAYER

Who governs governance? Rules accumulate. Complexity kills. Every failure adds another approval layer until nothing moves. The governance system itself must be governed.

NEW RULE [META-001]

"Rule Lifecycle Metadata": Every rule carries: RULE_ID, CREATED_DATE, OWNER (responsible for continued relevance), LAST_REVIEW_DATE, REVIEW_FREQUENCY (quarterly/semi-annually/annually), DEPENDENCIES (interacting rules), RETIREMENT_CRITERIA (per META-002). Not reviewed within frequency → "REVIEW_OVERDUE." Overdue by 2x → "POTENTIALLY_OBSOLETE" requiring urgent review. WHY: Rules without owners and review dates become permanent regardless of relevance.

NEW RULE [META-002]

"Rule Retirement Protocol": Every rule has defined retirement conditions: MVP-specific rules retire when trust progression relaxes them. Source-specific rules retire if source discontinued. Competitor-specific retire if competitor exits KSA. Timing-specific remain but only active in temporal window. Process: owner proposes with evidence, BD leadership approves, rule ARCHIVED not deleted (preserves audit trail), dependent rules reviewed for impact. WHY: System that only adds rules becomes incomprehensible. Retirement is hygiene.

NEW RULE [META-003]

"Governance Complexity Index" (GCI): $GCI = (Active\ Rules \times 1) + (Exceptions \times 3) + (Monthly\ Overrides \times 2) + (Rule\ Dependencies \times 1.5)$. Healthy: <500 MVP phase, <800 scale phase. Exceeds threshold → "COMPLEXITY_REVIEW": consolidate similar rules, eliminate exceptions by improving base rules, change rules instead of overriding, automate where trust permits. Target: reduce 10-15% per review without losing coverage. WHY: The day nobody can explain why the system did something is the day the system is ungovernable.

NEW RULE [META-004]

"Control Cost-Benefit": Every control must justify: $Cost = time\ hours/month + delay\ days + complexity\ increase$. $Risk\ Reduction = failure\ probability \times failure\ impact$. $Cost > 3x\ Risk\ Reduction$ → candidate for removal/simplification. Exception: CATASTROPHIC RISK controls (SEV-1 failures) maintained regardless — losing KSA market access exceeds any efficiency gain. Annual review: controls failing cost-benefit for 2 consecutive reviews → RETIRE. WHY: Organizations add controls after every failure but never remove them. Eventually controls cost more than failures they prevent.

NEW RULE [META-005]

"Dual Improvement Test": Every AI feature must answer before building: (1) How does it help humans decide better? (If not → automation for automation's sake) (2) How do humans improve it through usage? (If no feedback loop → will stagnate) (3) How is trust measured? (If unmeasurable → unmanageable) (4) How is accountability maintained? (If nobody responsible when wrong → unaccountable automation) (5) How does system recover if both AI and human fail? (If no recovery → fragile). Cannot answer all 5 → doesn't belong in production. May belong in research track. WHY: Prevents feature bloat adding complexity without decision quality.

ITERATION 7 — WHY HUMAN REVIEW EXISTS AT ALL

Review is NOT quality control. Review is ACCOUNTABILITY ASSIGNMENT. AI cannot suffer consequences. If Decimal loses SNB, AI loses nothing. Human loses relationship. Company loses opportunity. Every outbound action must have a responsible human.

NEW RULE [HUMANREVIEW-001]

"Accountability Attachment Record": Every outbound action stores: AI_RECOMMENDED_BY (chain version), REVIEWED_BY (human name + timestamp), ACCOUNT_OWNER (ultimately responsible human), DECISION (approve/edit/reject), REASONING (linked to HUMANDEC-001 taxonomy). Record is IMMUTABLE — cannot be edited after the fact. Answers: "When this message was sent to SNB's CEO, who approved it and why?" WHY: AI cannot appear in a meeting to explain why it sent a message. A human can. Accountability attaches to persons, not systems.

NEW RULE [HUMANREVIEW-002]

"Strategic vs Execution Review": STRATEGIC (higher stakes, less frequent): covers account targeting, stakeholder selection, outreach narrative, competitive positioning, timing strategy. Reviewer: owner or BD leadership (Level 2-3). Frequency: once per account per outreach cycle. EXECUTION (lower stakes, more frequent): individual message quality, wording, CTA. Reviewer: any certified (Level 1+). Frequency: per message during MVP, reducible per trust. CRITICAL: Strategic review is NEVER automated — even at FULL PARTNERSHIP, human always approves account STRATEGY. Execution review within approved strategic boundaries can auto-send if trust permits. WHY: Getting strategy wrong means EVERY message is wrong. This is the high-leverage review.

NEW RULE [HUMANREVIEW-003]

"Reasoning Checkpoint Access": Reviewer interface allows reviewing ANY intermediate step: expandable Agent 1 → 2 → 3 → 4 → 5 → 6 → 7. DEFAULT view: final message + strategy summary + signal reference (fast routine review). EXPANDED view: full chain (when something feels off). MANDATORY expansion: C-suite contact, SIGNAL_CONFLICT flag, ICS <60, first message to new account, Verifiability Score <50. WHY: Reviewing only the final message is reviewing the symptom. Reviewing reasoning is reviewing the cause.

NEW RULE [HUMANREVIEW-004]

"Reviewer Competency Tiers": Level 1 OPERATIONAL: P3/P4 to Director and below. Requires 4-week onboarding + certification (20 audited reviews, ≥90% agreement). Level 2 STRATEGIC: P1/P2, VP contacts, strategic reviews, conflicted accounts. Requires Level 1 experience (100+ reviews) + demonstrated strategic judgment + BD experience. Level 3 EXECUTIVE: C-suite messages, Domino Account outreach, market strategy, override approvals. Requires senior BD experience + deep KSA banking knowledge + multi-account relationship context. Messages routed to appropriate level. C-suite WAITS for Level 3 — better to delay than have junior reviewer approve CEO message. WHY: Not all reviews are equal. Not all reviewers are qualified for all reviews.

NEW RULE [HUMANREVIEW-005]

"Review Step Sunset Assessment": Every 6 months, per message category (signal priority x contact seniority x trust level): (a) VALUE_ADDED rate from HUMANDEC-005, (b) BLOCKED_HARM rate, (c) cost of this review step. VALUE_ADDED >20% OR BLOCKED_HARM >2% → JUSTIFIED, keep. VALUE_ADDED 5-20% AND BLOCKED_HARM <2% → QUESTIONABLE, consider spot-check (30% random). VALUE_ADDED <5% AND BLOCKED_HARM <1% → NOT JUSTIFIED, candidate for removal per trust progression. Exception: ANY category involving C-suite or P1 → ALWAYS justified regardless of metrics. WHY: Review that never changes outcomes is overhead. But some categories must stay reviewed because the consequence of the ONE error is too high.

ITERATION 8 — WHY AI REASONS THE WAY IT DOES

Good outcome does not equal good reasoning. Bad outcome does not equal bad reasoning. An AI that produces a meeting through lucky reasoning will produce a disaster when luck runs out. We must evaluate the PROCESS, not just the result.

NEW RULE [REASON-001]

"Reasoning-Outcome Independence": 4 quadrants: GOOD reasoning + GOOD outcome = IDEAL (system worked). GOOD reasoning + BAD outcome = UNLUCKY (good process, external factors — don't change reasoning, investigate external cause). BAD reasoning + GOOD outcome = LUCKY (DANGEROUS — creates false confidence, fix reasoning). BAD reasoning + BAD outcome = EXPECTED FAILURE (fix reasoning). After every significant outcome, evaluate each agent step: signal interpretation accurate? Research correct? Pain validated? Strategy appropriate? Produces "Reasoning Report Card" per outcome. WHY: Evaluate process, not just result.

NEW RULE [REASON-002]

"Critical Assumption Tracking": Every chain declares critical assumptions with confidence: Agent 1: "Signal is genuine (not PR). Signal is current. Event affects technology purchasing." Agent 2: "Strategic priorities are current. Technology stack accurate." Agent 4: "Inferred pains are real. Contact experiences these in their role." Agent 5: "Chosen angle will resonate. Competitive landscape is as understood." LOW-confidence assumptions highlighted in Reasoning Passport so reviewer knows where chain is weakest. WHY: Making assumptions explicit makes them challengeable. Hidden assumptions are unchallengeable — that's where failures hide.

NEW RULE [REASON-003]

"Evidence Strength Classification": Every claim tagged: FACT (HIGH): directly sourced from verified data. INFERENCE (MEDIUM): logically derived from facts. ASSUMPTION (LOW): reasonable but unverified role-based patterns. SPECULATION (VERY LOW): guess from minimal data. Rule: Messages must be 50%+ FACT or INFERENCE. Below 50% → "WEAK_EVIDENCE_CHAIN" flag — enrich before outreaching or caveat heavily. WHY: A message saying "I noticed your open banking project" when it's actually speculation is a hallucination-in-disguise.

NEW RULE [REASON-004]

"Inference Expiry Rules": Pain inferences: 60 days. Strategic priorities: 90 days. Persona motivations: 180 days (role-based, slower — BUT immediately invalidated if contact changes role). Competitive landscape: 60 days. Account research summary: 90 days. Stale input → "STALE_REASONING_INPUT" flag. Chain either refreshes first or proceeds with explicit caveat: "Pain inference based on data from X months ago. May not reflect current priorities." WHY: Using March's pain inference in September's message is navigating with an outdated map.

NEW RULE [REASON-005]

"Uncertainty Reduction as Purpose": Every step must quantify what uncertainty it reduced. Before Agent 1: "Signal detected, meaning unknown." After: "Digital lending investment likely. Buying intent: MODERATE-HIGH." After Agent 2: "Account is Tier 1 Islamic, maturity 4/10, priorities X, Y, Z." After Agent 4: "Most likely pain: onboarding friction, 2/5 App Store rating." Step that doesn't reduce uncertainty = "LOW_VALUE_STEP." Total chain still HIGH uncertainty = recommend INVESTIGATE (enrich/connector intel) instead of OUTREACH. WHY: Reasoning that doesn't reduce uncertainty is intellectual theater.

ITERATION 9 — DECISION INTELLIGENCE LAYER

The purpose of a recommendation is not to tell humans what to do. It is to help humans think better. A system that makes humans smarter is more valuable than one that makes them faster.

NEW RULE [DECISION-001]

"Counterfactual Display": Every recommendation shows next-best alternative: "RECOMMENDED: SNB (Effective Opportunity: 67). ALTERNATIVE: Riyadh Bank (62). SNB recommended because: stronger champion, higher regulatory urgency, KPMG connector. Riyadh Bank advantage: simpler committee, shorter procurement." WHY: Recommendation without alternatives is a directive. With alternatives is a decision aid. Human may see something in the alternative AI missed.

NEW RULE [DECISION-002]

"Decision Tradeoff Display": Every recommendation surfaces: Benefits ("SNB could yield largest KSA deal. Domino Account. KPMG path."), Risks ("Complex committee, 8+ stakeholders. Long procurement. High competitor entrenchment."), Opportunity Cost ("Focus on SNB = less attention to Bank AlJazira — simpler deal, faster close, lower strategic value."). WHY: Optimizing one dimension always costs another. Visible tradeoffs enable better judgment.

NEW RULE [DECISION-003]

"Ignored Recommendation Tracking": When human ignores/overrides: record recommendation, human alternative, stated reason (HUMANDEC-001), eventual outcome. Quarterly: "When humans overrode, were outcomes better (human right) or worse (AI right)?" Human right >60% → AI model needs updating. AI right >60% → investigate override biases. ~50/50 → genuinely ambiguous, present both in future. WHY: Systematic tracking reveals whether human overrides improve or degrade outcomes.

NEW RULE [DECISION-004]

"Decision vs Execution Quality Separation": Every outcome attributed to BOTH: GOOD DECISION + GOOD EXECUTION = success (replicate). GOOD DECISION + BAD EXECUTION = execution failure (fix process not strategy). BAD DECISION + GOOD EXECUTION = strategic error (fix model not process). BAD DECISION + BAD EXECUTION = total failure (fix both). After each cycle, owner classifies: was targeting/strategy right? Was execution good? WHY: Prevents blaming strategy for execution failures and vice versa.

NEW RULE [DECISION-005]

"Recommendation as Education": Every recommendation includes explanation improving understanding: NOT "Recommended: outreach Account X." BETTER: "Recommended: Account X because new CTO hire + Q1 budget timing + KPMG connector creates convergence historically correlating with high meeting rates. Key risk: competitor entrenchment in adjacent categories." Human finishes reading KNOWING MORE about enterprise sales. Over time, educational framing develops expertise, not compliance. WHY: System making humans smarter is more valuable than one making them faster. Speed without understanding is fragile.

ITERATION 10 — COGNITIVE DIVISION OF LABOR

The most dangerous hidden assumption: 'humans are smarter than AI' (false) and 'AI is smarter than humans' (also false). The real question: what kinds of decisions should each system make? AI sees 1000 accounts. Human sees 1 relationship. AI optimizes information. Humans optimize reality. The real output of the engine is not messages or meetings. It is DECISION CONFIDENCE.

NEW RULE [COGNITIVE-001]

"Information Boundary Rule": AI decides autonomously ONLY within information boundary (data system possesses). Human required whenever off-system context materially affects outcome. Off-system context: personal relationships (dinner with CTO last week), political situations (CDO rumored to be leaving), confidential info (connector shared NDA-protected detail verbally), cultural timing (family event in executive's community), market gossip (bank about to be acquired). Test: "Could information OUTSIDE system data change this decision?" If plausible → human involved. WHY: AI is omniscient within its database and completely blind outside it. Humans are the opposite.

NEW RULE [COGNITIVE-002]

"Pattern vs Exception Allocation": AI handles PATTERNS: "This signal type converts at X rate. Accounts with this profile respond to this angle. This persona prefers these proof points." Humans handle EXCEPTIONS: "This bank never responds to cold email. This CTO only accepts KPMG introductions. This account is about to be acquired — all signals are noise." AI patterns are DEFAULT. Human-identified exceptions OVERRIDE for specific account/contact. Exceptions don't change the pattern (one exception doesn't prove pattern wrong). They create LOCAL OVERRIDE persisting until reviewed. WHY: Patterns are right MOST of the time. Exceptions are right for SPECIFIC situations. System needs both.

NEW RULE [COGNITIVE-003]

"Mental Model Capture": When human overrides AI with expert judgment, system prompts: "What principle informed your decision?" Response options: INSTITUTIONAL_BEHAVIOR ("This bank always does X"), RELATIONSHIP_DYNAMIC ("This person responds to Y"), MARKET_PATTERN ("In this segment, Z is happening"), POLITICAL_READ ("Internal dynamics suggest W"), TIMING_INTUITION ("The timing feels wrong because..."). Captured mental models become Decision Pattern Library entries after 3+ occurrences across different situations. WHY: Expert mental models are the most valuable human-generated data. Without capture, they leave when experts leave.

NEW RULE [COGNITIVE-004]

"Disagreement as Signal": AI HIGH + Human LOW → INVESTIGATE (AI sees something human doesn't, or vice versa — surface both reasonings). Human HIGH + AI LOW → INVESTIGATE (often reveals dark intent or relationship context not in system). BOTH HIGH, DIFFERENT conclusions → HIGHEST VALUE DISAGREEMENT (both valid, escalate to leadership for strategic arbitration, log resolution as Decision Pattern). BOTH LOW → DEFER (neither has enough info, prioritize INVESTIGATE and ENRICH). WHY: Most systems treat disagreement as problem. This system treats it as the most informative signal available. Alignment tells you nothing. Disagreement tells you where your model is incomplete.

NEW RULE [COGNITIVE-005]

"Decision Output Format": Every major recommendation outputs three components: CONFIDENCE: "We are X% confident this is right." (from ICS, evidence strength, model reliability). UNCERTAINTY: "Key unknowns are: [specific gaps]." (What information gaps remain?) NEXT BEST QUESTION: "The single piece of information that would most improve confidence is: [specific question]." Example: "CONFIDENCE: 65% pursue SNB this quarter. UNCERTAINTY: budget status unknown, 3 of ~7 stakeholders identified, competitor entrenchment unclear. NEXT BEST QUESTION: Can KPMG connector confirm whether SNB budgeted for lending tech in 2027?" WHY: Confidence without uncertainty is false precision. Uncertainty without next step is paralyzing. NEXT BEST QUESTION is the most actionable output.

NEW RULE [PRINCIPLE-3-REWRITE]

"Final Principle Definition": The ABM Engine is a distributed decision-making system. AI contributes memory, analytical scale, pattern recognition, consistency, and uncertainty reduction across the account universe. Humans contribute accountability, contextual judgment, political intelligence, relationship stewardship, ethical reasoning, and exception handling. The architecture does not seek to maximize automation. It seeks to maximize decision quality under uncertainty while preserving trust, accountability, explainability, resilience, and human expertise. Every component must answer: What uncertainty does this reduce? Which cognitive architecture is best suited? What is the cost of being wrong? How is trust maintained? How does the system recover if both fail? WHY: This transforms design philosophy from "AI tool with approval button" to "enterprise decision intelligence system."

PRINCIPLE 3 — COMPLETE RULE INDEX

Section	Rules	Count
Initial Pass (HUMAN-AUTO)	001-004	4
Foundation & Context	AIHUMAN-FOUNDATION/CONTEXT/DISAGREE	5
Explainability	EXPLAIN-001-005	5
AI Boundaries	AIBOUNDARY-001-003	3
FM1: Rubber-Stamping	FAIL-RUBBER-001-002	2
FM2: Hallucination	FAIL-HALLUC-001-002	2
FM3: AI Staleness	FAIL-STALE-001-002	2
FM4: Over-Reliance	FAIL-OVERRELY-001	1
FM5: Wrong Metrics	FAIL-WRONGMETRIC-001-002	2
FM6: Memory Leakage	FAIL-LEAK-001-002	2
Trust Progression	TRUST-001	1
Iter 1: AI Learning	AILEARN + AIMODEL + AIQUALITY	10
Iter 2: Human Capability	HUMANSKILL + FATIGUE + TRANSPARENCY + MEMORY	9
Iter 3: Adversarial/Catastrophic/Ethical	ADVERSARIAL + CATASTROPHE + ETHICS + GOVERNANCE	14
Iter 4: Trust Economics	AITRUST-001-005 + 001a	6
Iter 5: Decision Science	HUMANDEC-001-005	5
Iter 6: Meta-Governance	META-001-005	5
Iter 7: Why Review Exists	HUMANREVIEW-001-005	5
Iter 8: Why AI Reasons	REASON-001-005	5
Iter 9: Decision Intelligence	DECISION-001-005	5
Iter 10: Cognitive Division	COGNITIVE-001-005 + REWRITE	6

TOTAL: 104 RULES

From 'AI assists humans' to a complete Distributed Decision Intelligence Architecture.
10 iterations of 5-Whys excavation.
Every rule has full explanation, rationale, and implementation specifics.
Ready for technical implementation.

END OF PRINCIPLE 3 DOCUMENT

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